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Greenwood et al.

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(54) **BRAIDED IMPLANT WITH DETACHMENT MECHANISM**

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(65) **Prior Publication Data**

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(51) **Int. Cl.**
A61B 17/12 (2006.01)
A61B 17/00 (2006.01)

(52) **U.S. Cl.**
CPC .. ***A61B 17/12113*** (2013.01); ***A61B 17/12031*** (2013.01); ***A61B 17/12168*** (2013.01); *A61B 2017/00915* (2013.01); *A61B 2017/12054* (2013.01)

(58) **Field of Classification Search**
CPC A61B 17/12113; A61B 17/12031; A61B 17/12168; A61B 17/17172; A61B 2017/00915; A61B 2017/12054; A61B 2017/00526
See application file for complete search history.

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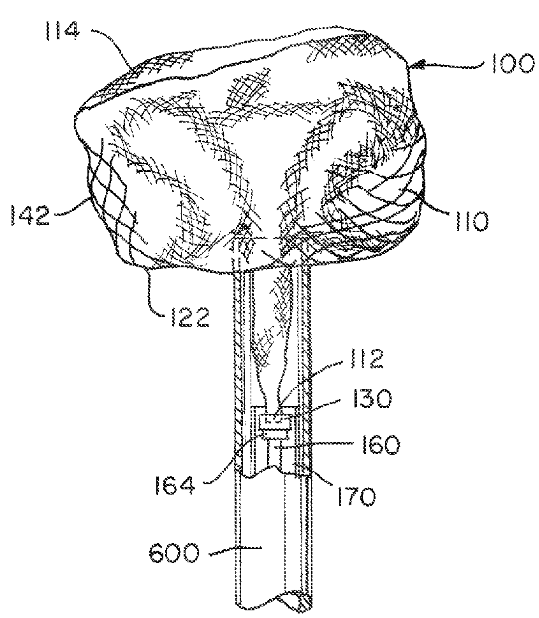
(74) *Attorney, Agent, or Firm* — TROUTMAN PEPPER LOCKE LLP

(57) **ABSTRACT**

An occlusive device can include a segment including an open end and a proximal end. The occlusive device can include a push wire that is positioned proximal of the proximal end. The occlusive device can include a detachment feature that is attached to the push wire via a detachment groove. The occlusive device can include a sleeve surrounding the push wire. In a non-deployed configuration, the sleeve can surround the detachment feature and prevent release of the proximal end of the segment from the push wire. In a deployed configuration, the proximal end can be flush with a proximal end of the spherical cavity.

20 Claims, 7 Drawing Sheets



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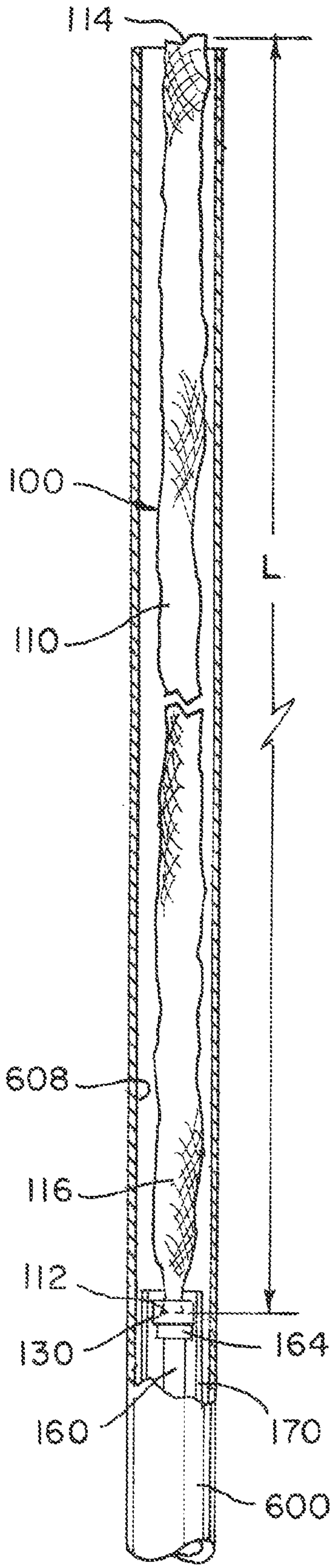
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600

FIG. 1A

FIG. 1B

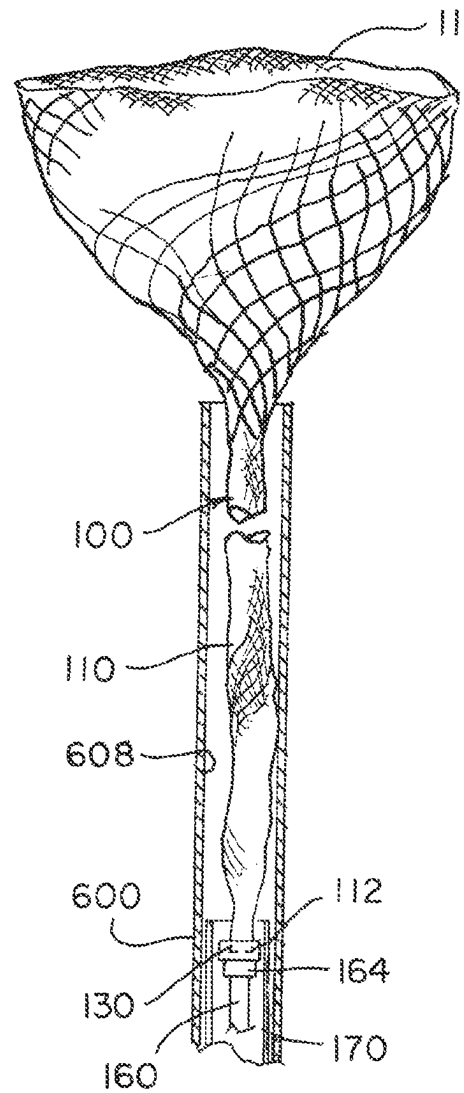
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FIG. 1C

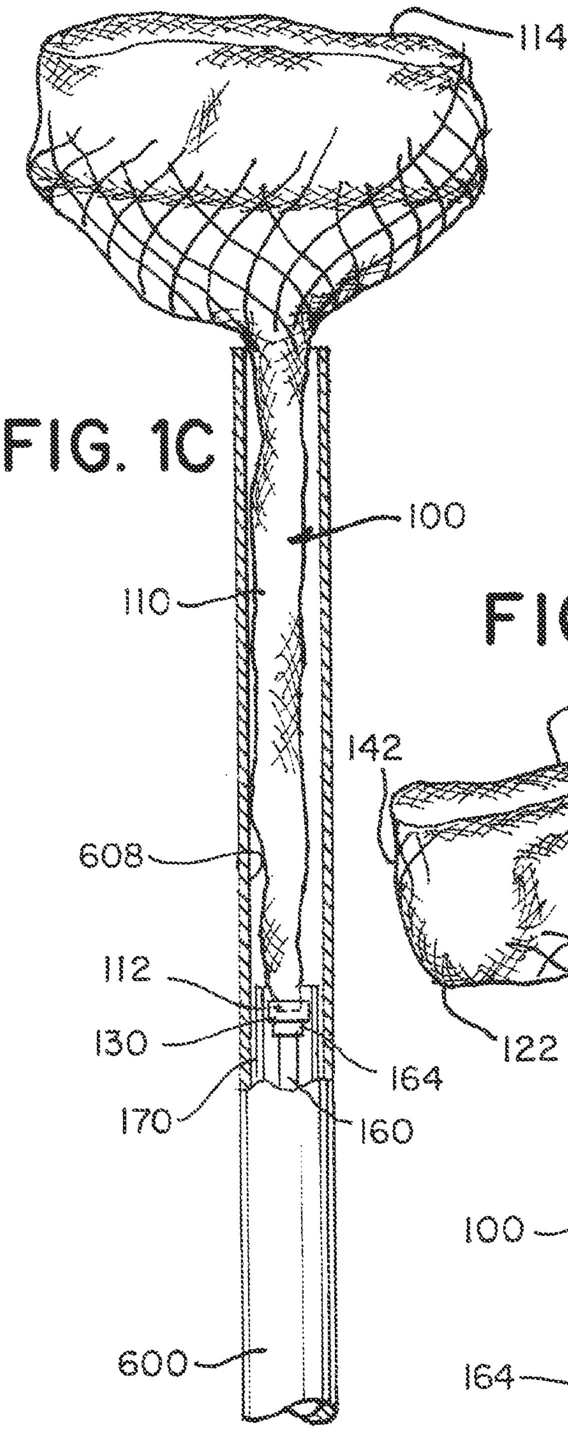


FIG. 1D

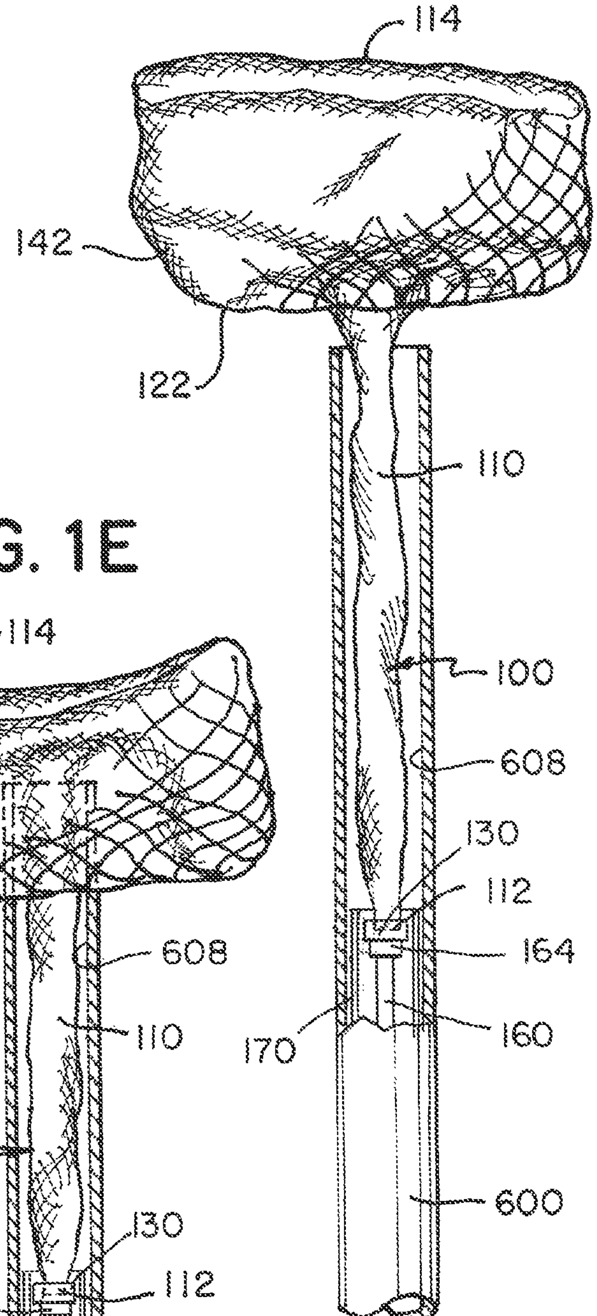
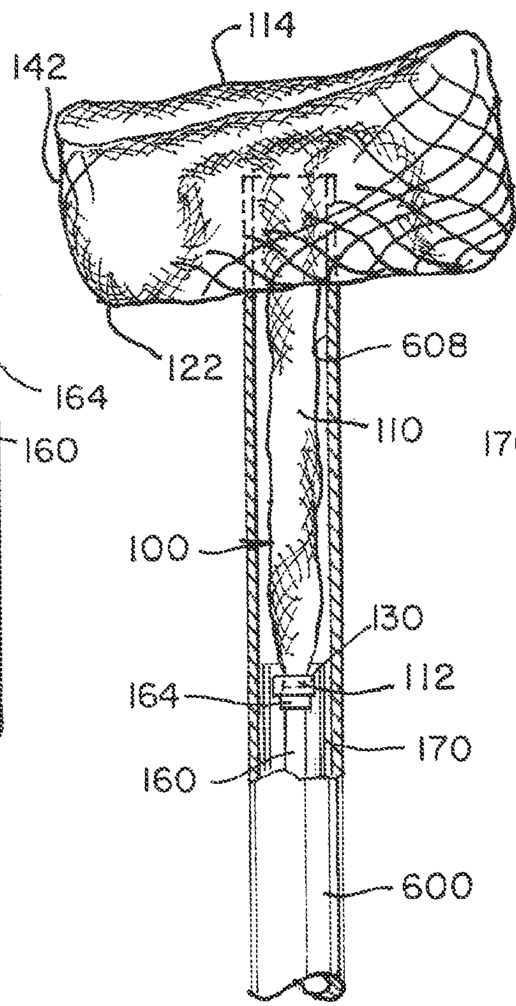


FIG. 1E



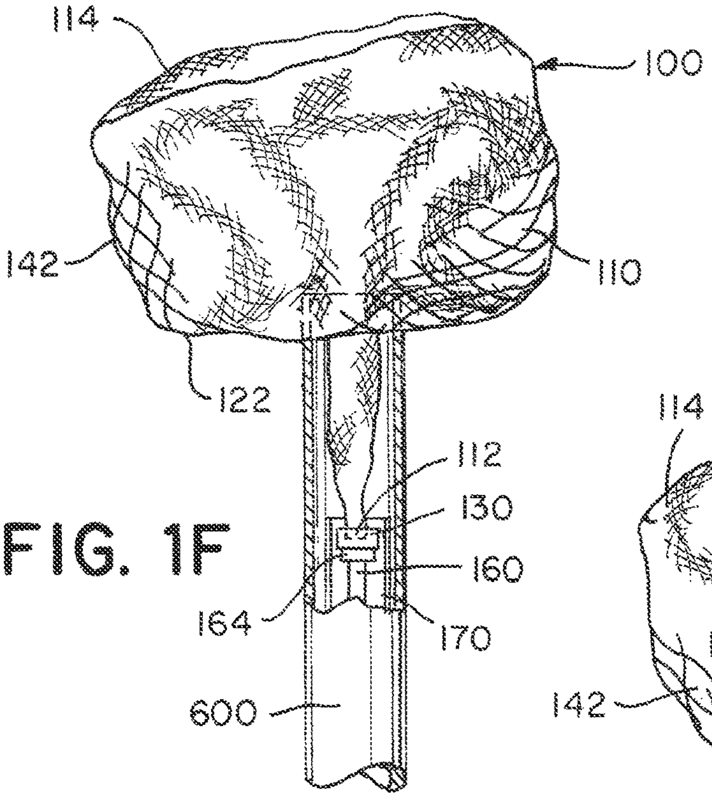


FIG. 1F

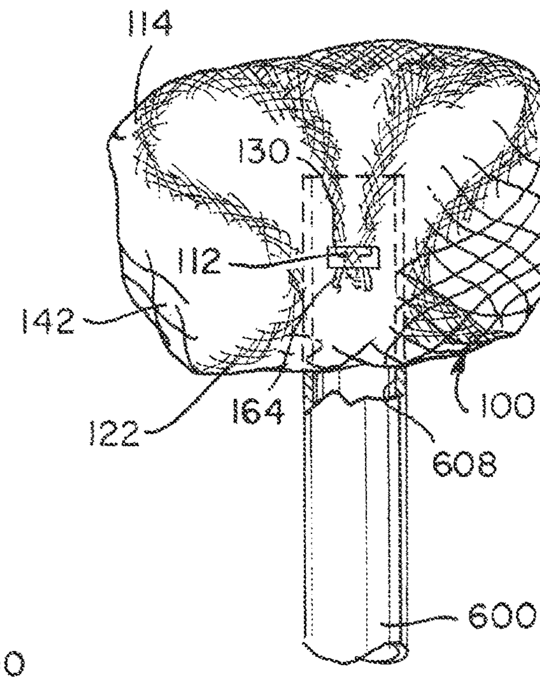


FIG. 1G

FIG. 2A

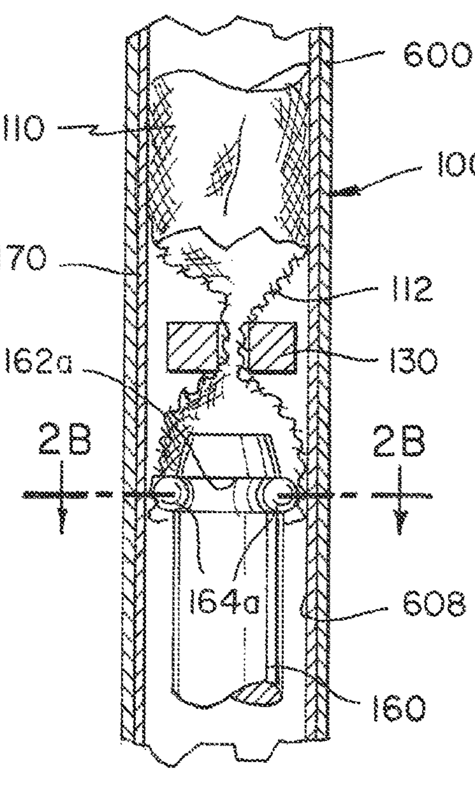
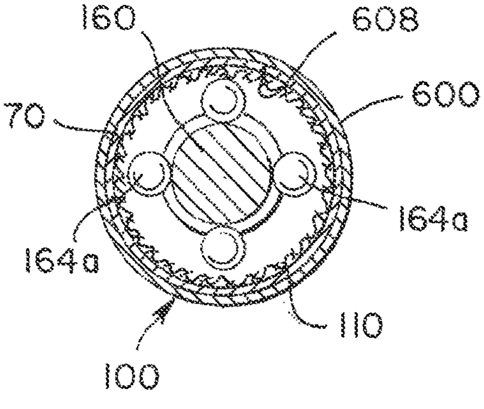


FIG. 2B



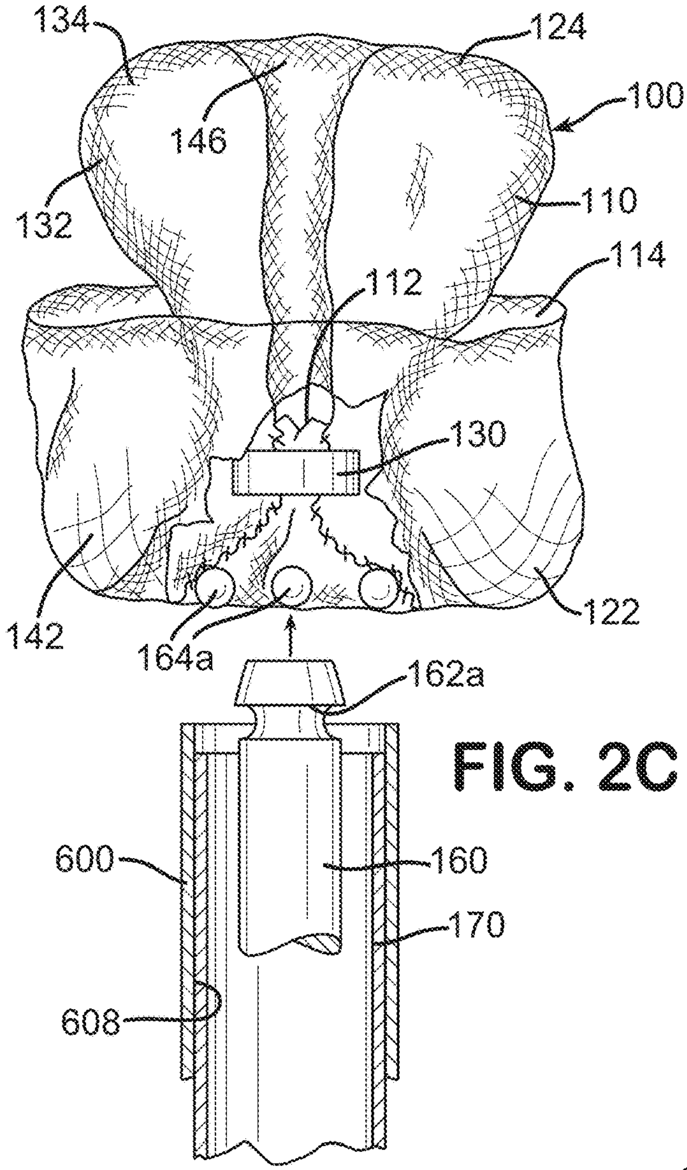


FIG. 2C

FIG. 3A

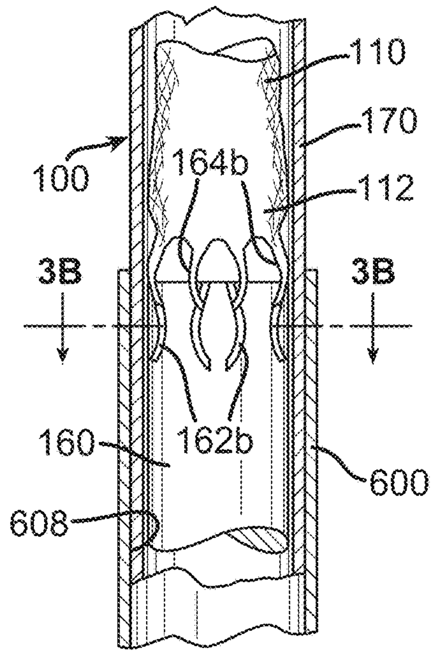


FIG. 3C

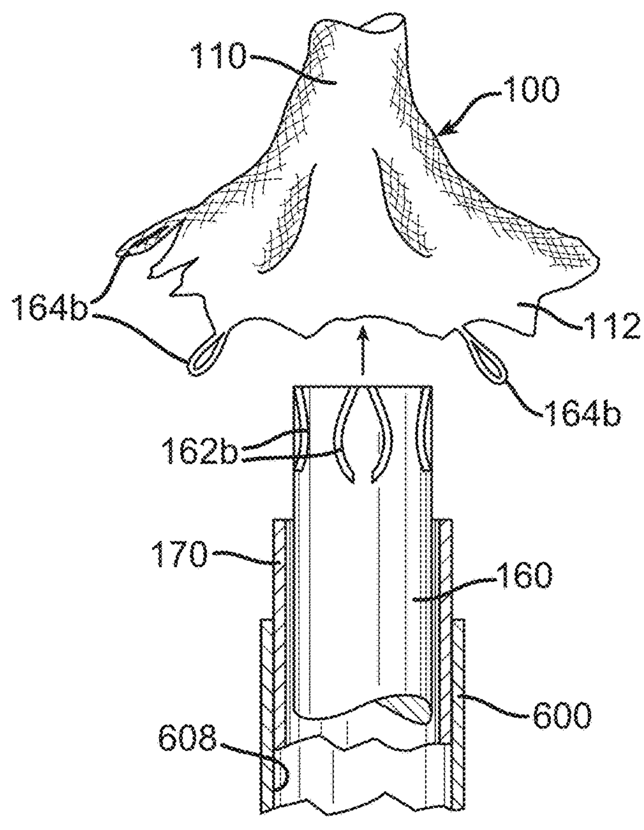


FIG. 3B

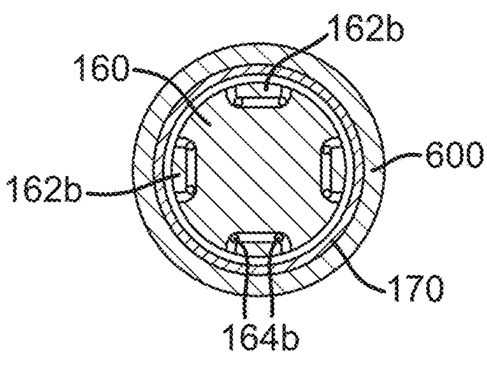
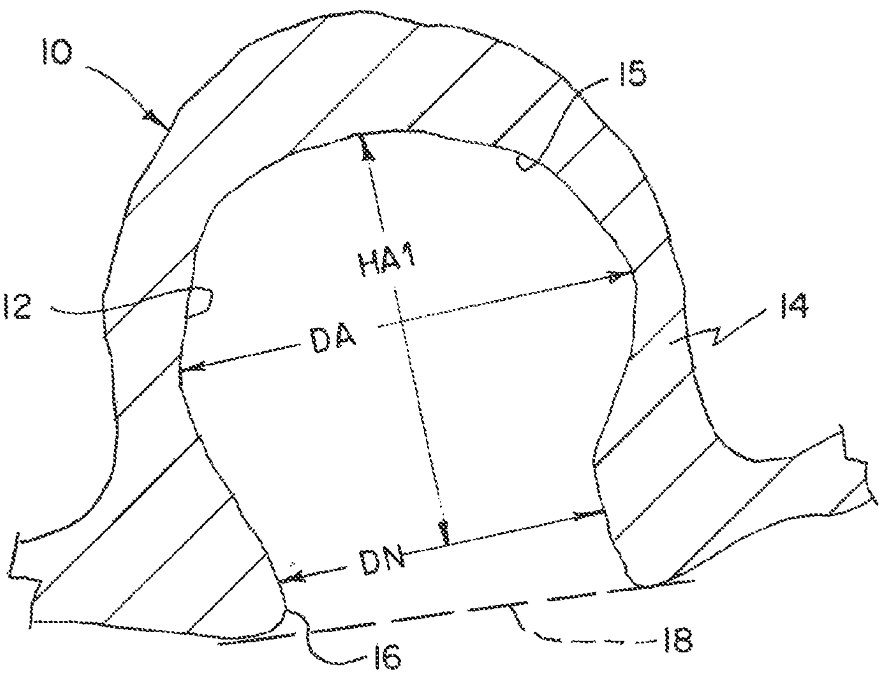


FIG. 4A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HA1
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DA
14
DN
16
18

FIG. 4B

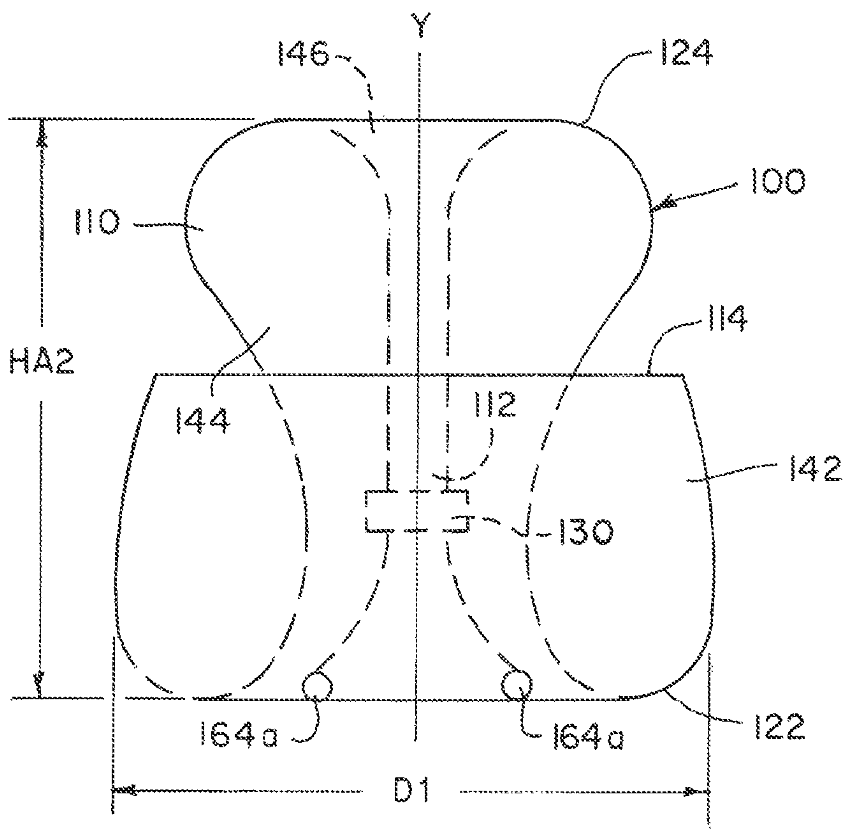

Y
146
124
100
110
114
HA2
144
112
142
130
164a
164a
122
D1

FIG. 5

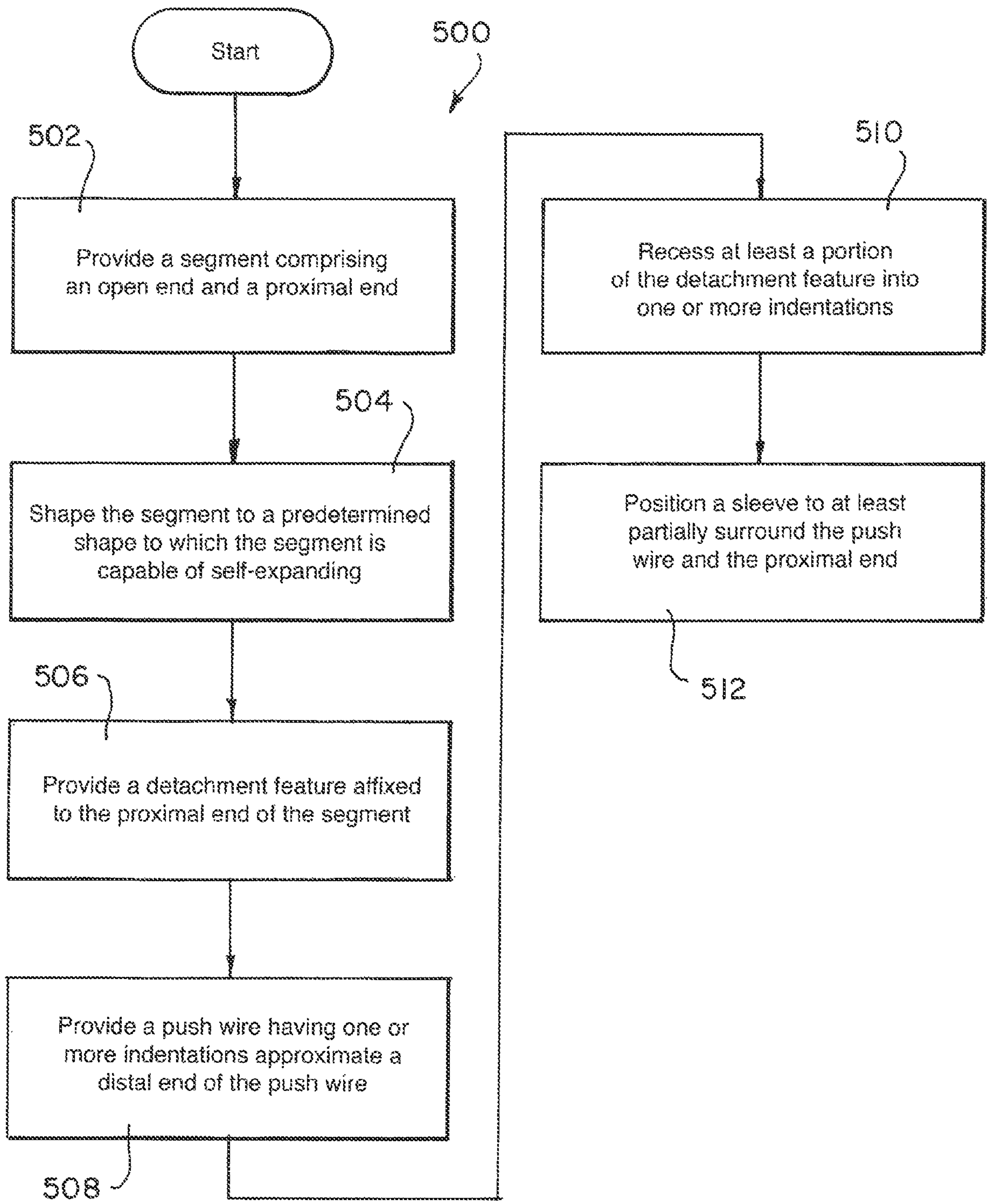
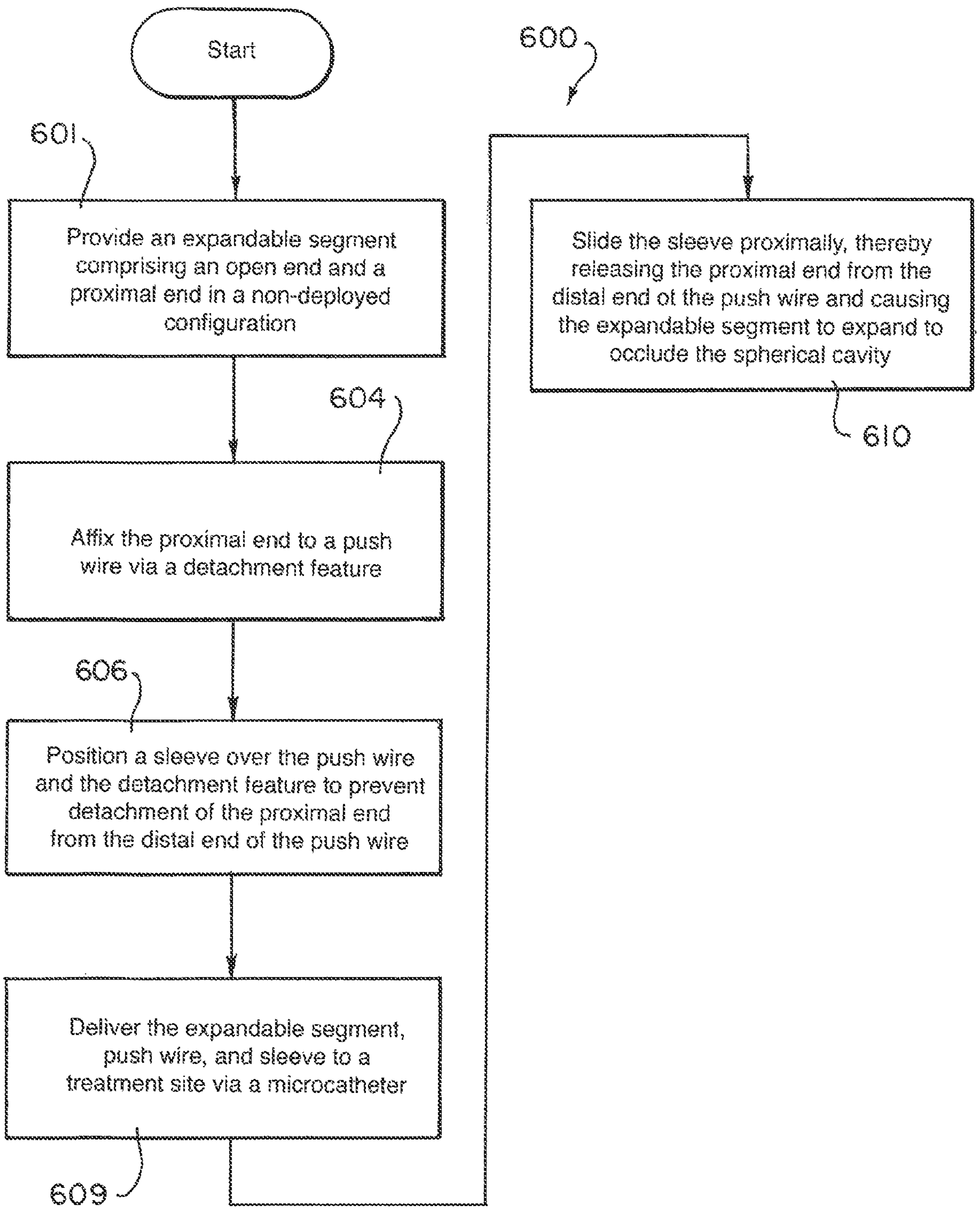

Start
500
502
Provide a segment comprising an open end and a proximal end
504
Shape the segment to a predetermined shape to which the segment is capable of self-expanding
506
Provide a detachment feature affixed to the proximal end of the segment
Provide a push wire having one or more indentations approximate a distal end of the push wire
508
510
Recess at least a portion of the detachment feature into one or more indentations
Position a sleeve to at least partially surround the push wire and the proximal end
512

FIG. 6


Start
600
601
Provide an expandable segment comprising an open end and a proximal end in a non-deployed configuration
604
Affix the proximal end to a push wire via a detachment feature
606
Position a sleeve over the push wire and the detachment feature to prevent detachment of the proximal end from the distal end of the push wire
Deliver the expandable segment, push wire, and sleeve to a treatment site via a microcatheter
609
Slide the sleeve proximally, thereby releasing the proximal end from the distal end of the push wire and causing the expandable segment to expand to occlude the spherical cavity
610

BRAIDED IMPLANT WITH DETACHMENT MECHANISM

FIELD OF INVENTION

The present invention generally relates to medical instruments, and more particularly, to embolic implants detachment mechanisms for aneurysm therapy.

BACKGROUND

Cranial aneurysms can be complicated and difficult to treat due to their proximity to critical brain tissues. Recently, tubular braided implants have been introduced that have the potential to treat an aneurysm or other arterio-venous malformation easily, accurately, and safely in a parent vessel without blocking flow into perforator vessels communicating with the parent vessel. Implant devices for treating aneurysms must be delivered through long, small, tortuous blood vessels and positioning must be controlled precisely to ensure aneurysm filling without causing additional occlusions or clotting in nearby vessels. Accordingly, it is necessary to have a delivery and detachment mechanism providing the connection point between a tubular braided implant and a delivery catheter that has the ability to deliver, position, manipulate, and then release the implant.

SUMMARY

It is an object of the present invention to provide systems, devices, and methods to meet the above-stated needs. Generally, it is an object of the present invention to provide an endovascular implantation system for occluding a spherical cavity. The endovascular implantation system can include an occlusive segment that includes an open end and a proximal end and is configured to occlude a spherical cavity. The endovascular implantation system can include a push wire that is positioned proximal of the proximal end of the occlusive segment. The push wire can include one or more indentations on an outer surface of the push wire approximate a distal end of the push wire. The proximal end of the occlusive segment can include a detachment feature that is shaped to at least partially recess into the one or more indentations of the push wire. The occlusive device can include a sleeve surrounding at least a portion of the push wire. In a non-deployed configuration, the sleeve can at least partially surround the detachment feature and prevent release of the proximal end of the occlusive segment from the push wire. In a deployed configuration, the sleeve can be retracted proximally from the detachment feature to thereby detach the occlusive segment from the push wire.

In some examples, the detachment feature can further include a plurality of wire loops that extend from the proximal end of the segment that are configured to fit into the one or more indentations that are positioned on a distal tip of the push wire.

In some examples, the one or more indentations can include a plurality of laser-etched pockets. Each of the laser-etched pockets can be shaped to fit a corresponding wire loop of the plurality of wire loops.

In some examples, the detachment feature can further include a plurality of balls that extend from the proximal end of the segment. The plurality of balls can be configured to fit into the one or more indentations positioned on distal tip of the push wire.

In some examples, the one or more indentations can include an annular groove circumscribing an outer surface of the push wire.

In some examples, in the deployed configuration, the occlusive segment can expand to occlude the spherical cavity.

In some examples, in the deployed configuration the occlusive segment can extend in a distal direction from the proximal end and can include two inversions that separate three sections of the segment which at least partially overlap each other such that the proximal end is affixed to an innermost section of the three sections and a middle section of the three sections can extend between the two inversions and is positioned within an outermost section and around the innermost section.

In some examples, the endovascular implantation system can include a microcatheter that is configured to deliver the occlusive segment, the push wire, and the sleeve to the spherical cavity with the occlusive device in the non-deployed configuration. The microcatheter can include a lumen, and the segment can have a diameter in the non-deployed configuration sized to fit within the lumen of the microcatheter.

In some examples, the endovascular implantation system can include a band that is approximate the proximal end of the segment. The band can include radiopaque material.

In some examples, in the deployed configuration, the open end of the occlusive segment can be positioned approximate a distal wall of the spherical cavity and the band can be suspended within the spherical cavity.

In some examples, the spherical cavity can be an aneurysm sac.

In some examples, the occlusive segment can include a tubular braid.

In another aspect, a method for constructing an endovascular treatment system is disclosed. The method can include providing an occlusive segment that includes an open end and a proximal end. The method can include shaping the occlusive segment to a predetermined shape to which the segment is capable of self-expanding. Shaping the predetermined shape can include inverting the segment to form a proximal inversion that is folded towards the distal direction to thereby define an outermost section of the segment. Shaping the predetermined shape can include inverting the segment to form a distal inversion folded toward the proximal direction to thereby define a middle section between the proximal and distal inversion of the segment that is at least partially surrounded by the outermost section and defines an innermost section between the distal inversion and the proximal end that is at least partially surrounded by the middle section. The method can include providing a detachment feature that is affixed to the proximal end of the segment. The method can include providing a push wire having one or more indentations in a distal end of the push wire. The method can include recessing at least a portion of the detachment feature into the one or more indentations of the push wire. The method can include positioning a sleeve to at least partially surround the push wire and the detachment feature.

In some examples, the sleeve is effective to retain the proximal end of the segment attached to the push wire.

In some examples, the detachment feature can further include a plurality of wire loops that extend from the proximal end that can be configured to fit into the one or more indentations positioned on the distal end of the push wire.

In some examples, the one or more indentations can include a plurality of laser-etched pockets. Each of the laser-etched pockets can be shaped to fit a corresponding wire loop of the plurality of wire loops.

In some examples, the detachment feature can further include a plurality of balls that extend from the proximal end of the segment. The plurality of balls can be configured to fit into the one or more indentations that are positioned on the distal end of the push wire.

In some examples, in the deployed configuration, the proximal end of the segment can be flush with a proximal end of the spherical cavity. In some examples, the spherical cavity can be an aneurysm sac.

In another aspect, a method of occluding a spherical cavity is disclosed. The method can include providing an expandable segment that includes an open end and a proximal end in a non-deployed configuration. The method can include affixing the proximal end to a push wire via a detachment feature. The method can include positioning a sleeve over at least a portion of the push wire and at least a portion of the detachment feature to prevent detachment of the proximal end of the expandable segment from the distal end of the push wire. The method can include delivering the expandable segment, push wire, and sleeve to a treatment site via a microcatheter. The method can include sliding the sleeve proximally, thereby releasing the proximal end from the distal end of the push wire and causing the expandable segment to expand to a deployed configuration to occlude the spherical cavity.

In some examples, the detachment feature can further include a plurality of wire loops that extend from the proximal end that can be configured to fit into the one or more indentations positioned on the distal end of the push wire.

In some examples, the one or more indentations can include a plurality of laser-etched pockets. Each of the laser-etched pockets can be shaped to fit a corresponding wire loop of the plurality of wire loops.

In some examples, the detachment feature can further include a plurality of balls that extend from the proximal end of the segment. The plurality of balls can be configured to fit into the one or more indentations that are positioned on the distal end of the push wire.

In some examples, the one or more indentations can include an annular groove circumscribing an outer surface of the push wire.

In some examples, the proximal end is flush with a proximal end of the spherical cavity. In some examples, the spherical cavity can be an aneurysm.

In some examples, in the deployed configuration, the expandable segment can include two inversions that separate three sections which at least partially overlap each other such that the proximal end is affixed to an innermost section of the three sections, and a middle section of the three sections extends between the two inversion and is positioned within an outermost section and around the innermost section.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and further aspects of this invention are further discussed with reference to the following description in conjunction with the accompanying drawings, in which like numerals indicate like structural elements and features in various figures. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating principles of the invention. The figures depict one or more implementations of the inventive devices, by way of example only, not by way of limitation.

FIGS. **1**A through **1**G are illustrations of an example endovascular treatment system including an implant having a tubular braid that expands to a predetermined shape as the tubular braid exits a microcatheter according to aspects of the present invention;

FIG. **2**A is an illustration of an example endovascular treatment system including an implant with a detachment feature that includes balls in a non-deployed configuration, according to aspects of the present invention;

FIG. **2**B is cross-sectional illustration of the system as indicated in FIG. **2**A according to aspects of the present invention;

FIG. **2**C is an illustration of the system of FIG. **2**A with the detachment feature in a deployed configuration, according to aspects of the present invention;

FIG. **3**A is an illustration of another example endovascular treatment system including an implant with a detachment feature including loops and a push wire including laser-etched pockets in a non-deployed configuration, according to aspects of the present invention;

FIG. **3**B is a cross-sectional illustration of the system of FIG. **3**A according to aspects of the present invention;

FIG. **3**C is an illustration of the system of FIG. **3**A with the detachment feature in a deployed configuration, according to aspects of the present invention;

FIGS. **4**A and **4**B are illustrations of measurements of an example spherical cavity and implant according to aspects of the present invention;

FIG. **5** is a flowchart of a method for constructing an endovascular treatment system, according to aspects of the present invention; and

FIG. **6** is a flowchart of a method for occluding a spherical cavity, according to aspects of the present invention.

DETAILED DESCRIPTION

As used herein, the terms "about" or "approximately" for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein. More specifically, "about" or "approximately" may refer to the range of values ±20% of the recited value, e.g. "about 90%" may refer to the range of values from 71% to 99%.

When used herein, the terms "tubular" and "tube" are to be construed broadly and are not limited to a structure that is a right cylinder or strictly circumferential in cross-section or of a uniform cross-section throughout its length. For example, the tubular structure or system is generally illustrated as a substantially right cylindrical structure. However, the tubular system may have a tapered or curved outer surface without departing from the scope of the present invention.

Examples presented herein generally include a braided implant that can secure within an aneurysm sac and occlude a majority of the aneurysm's neck. The implant can include a tubular braid that can be set into a predetermined shape, compressed for delivery through a microcatheter, and implanted in at least one implanted position that is based on the predetermined shape and the geometry of the aneurysm in which the braid is implanted. When compressed, the implant can be sufficiently short to mitigate friction forces produced when the implant is delivered unsheathed through the microcatheter allowing for a more simplistic delivery system compared to some other known braided embolic implant delivery systems. The implant can be as described in U.S. Pat. No. 10,653,425, the entirety of which is incorporated herein by reference as if included in full.

FIGS. **1**A through **1**G are illustrations of an example implant **100** having a segment **110** expanding to a deployed configuration as the segment **110** exits a microcatheter **600**. The implant **100** can have a deployed shape similar to as illustrated below in reference to FIGS. **1**G, **2**C and **4**B. As illustrated in FIG. **1**A, the segment **110** can be shaped to a non-deployed shape that is extended to a single layer of tubular braid having a compressed circumference/diameter sized to be delivered through the microcatheter **600** and a length L. The illustrated implant **100** can have a length L of between about 22 mm and about 25 mm. As will be appreciated and understood by a person skilled in the art, the length L of a specific segment **110** can be tailored based on the size and shape of the aneurysm being treated.

During delivery through the microcatheter **600**, the detachment feature **164** can be attached to a delivery system at a proximal end **112** of the implant **100**, and the open end **114** can define the distal end of the implant **100**. The delivery system is drawn simplistically in FIGS. **1**A through **1**C for the sake of simplicity of illustration, and example delivery system configurations are shown in greater detail in FIGS. **2**A through **2**C and **3**A through **3**C. Collapsing the segment **110** to a single layer tube can result in a segment **110** that has a sufficiently small diameter and a sufficiently short length L to mitigate effects of friction force on the segment **110** when it is delivered through lumen **608** of the microcatheter **600**, allowing the segment **110** to be delivered unsheathed in some applications. In some applications, segment **110** is delivered through lumen **608** of microcatheter **600** with a sheath **170** covering some or all of segment **110**. Sheath **170** can be configured to be placed over at least detachment feature **164** to keep detachment feature **164** from releasing the connection between a distal end of push wire **160** and the implant **110**.

As illustrated in FIG. **1**B, the open end **114** can be positioned to exit the microcatheter **600** before any other portion of the segment **110** exits the microcatheter. The open end **114** can expand as it exits the microcatheter **600**. If the open end **114** is unconstrained by an aneurysm as illustrated, the open end can expand to its circumference in the predetermined shape.

As illustrated in FIG. **1**C, the distal portion of the segment **110** can continue to expand radially as it exits the microcatheter **600**.

As illustrated in FIG. **1**D, the segment **110** can form an inversion **122** defining an outer section **142** as the segment **110** is further pushed out of the microcatheter **600**.

As illustrated in FIGS. **1**E, **1**F, and **1**G, the "S" shape of a middle section **144** can begin to form as the segment **110** is further pushed from the microcatheter **600**.

FIG. **2**A is an illustration of an example endovascular treatment system **100** including an implant with a detachment feature, in a non-deployed configuration, that includes balls, according to an exemplary embodiment. FIG. **2**B is cross-sectional illustration of the system **100** as indicated in FIG. **2**A. As shown, implant **100** can be delivered through a lumen **608** of microcatheter **600** to a treatment site. The implant can include a segment **110** which can be formed as a heat-set tubular braided implant that, when released from the microcatheter at a treatment site, the segment **110** can expand into a deployed configuration as shown in FIG. **2**C. The segment can include an open end **114** (not shown) and a proximal end **112** that can be attached to a push wire **160** via a detachment feature **164**A. Detachment feature **164**A can include one or more balls that are shaped to fit into one or more indentations **162**A located on distal end of push wire **160**. The one or more indentations **162** can include an annular groove circumscribing the outer surface of the push wire **160** that is sized to receive the balls. Additionally, or alternatively, the one or more indentations **162** can include a plurality of circular or semi-spherical pockets sized to receive the balls. A sheath **170** can be positioned over the implant **100** such that sheath **170** covers at least a portion of the detachment feature **164**A, thereby retaining the detachment feature **164**A at least partially recessed within the one or more indentations **162**A until the sheath **170** is removed. As illustrated, approximate the proximal end of the segment **110** can be a band **130**. However, band **130** is an optional feature and implant **100** can be configured to not include band **130**. Band **130** can be made of a radiopaque material in order to facilitate tracking the position of implant **100** as implant **100** is delivered through vasculature to a treatment site.

FIG. **2**C is an illustration of the system of FIG. **2**A in a deployed configuration. After implant **100** is delivered to a treatment site (e.g., spherical cavity **10**, as described below in FIG. **4**A), the implant **100** can be manipulated to expand from the non-deployed configuration to the deployed configuration by pushing the implant **100** out from microcatheter **600**. FIG. **2**C shows the implant **100** detached from push wire **160**. The sheath **170** has been removed by being slid proximally into microcatheter **600**, thus allowing detachment feature **164**A to detach from the one or more indentations **162**A. In some examples, there can be one detachment feature **164**A and one corresponding indentation or groove **162**A, two detachment features **164**A and two corresponding indentations or grooves **162**A, or more than two detachment features **164**A and corresponding indentations or grooves **162**A. As shown, implant **100** can expand to the deployed configuration having two inversions **122**, **124**, dividing the segment **110** into three segments **142**, **144**, **146**. In the deployed configuration, the segment **110** can have an outer section **142** extending from the open end **114** of the segment **110** to one of the inversions **122**, an inner section **146** extending from the proximal end **112** of the segment **110** to the other of the inversions **124**, and a middle section **144** extending between the two inversions **122**, **124**.

FIG. **3**A is an illustration of another example endovascular treatment system including an implant with a detachment feature including loops and a push wire including laser-etched pockets in a non-deployed configuration. Proximal end **112** of implant **100** segment **110** is shown with sheath **170** extended over the proximal end **112** in FIG. **3**A. Sheath **170** can be translated proximally, thereby exposing the proximal end **112** of segment **110** and allowing detachment feature **164**B to be released from corresponding grooves **162**B. The detachment feature **164**B can be one or more loops formed on the proximal end **112** of the implant segment **110**. Push wire **160** can include one or more indentations **162**B on the distal end of push wire **160**. In some examples, the one or more indentations **162**B can include laser-etched pockets that are shaped to fit the one or more loops of detachment feature **164**B.

FIG. **3**B is a cross-sectional view of the system **100** showing the loop and pocket detachment mechanism as indicated in FIG. **3**A. As can be seen in FIG. **3**B, the pockets **162**B can extend radially into the push wire **160** and correspond to the shape of the one or more loops of detachment feature **164**B.

FIG. **3**C is an illustration of the system **100** after the detachment feature **164***b* expands so that loops release from the indentations **162**B of the push wire **160**. The sheath **170** is retracted to allow the detachment feature **164***b* to expand.

FIGS. **4**A and **4**B are illustrations of measurements of an example spherical cavity and implant according to aspects of the present invention. FIG. **4**A is an illustration of height HAL sac diameter DA, and neck diameter DN, which are measurements of a spherical cavity **10**. In some examples, spherical cavity can be an aneurysm of a patient. The location of the plane **18** defining a boundary between the spherical cavity **10** and blood vessels is also illustrated. As shown, the spherical cavity **10** can include a neck **16**, a wall **14**, and a distal wall **15**. The spherical cavity **10** can also include a spherical cavity interior **12**. The endovascular implant **100** can be configured to occlude the spherical cavity **10** when the implant is in a deployed configuration.

FIG. **4**B is an illustration of height HA**2** of an example implant **100** in a deployed configuration. In the deployed configuration, the braid **110** of the example implant **100** can be substantially radially symmetrical about vertical axis y, and therefore can have substantially circular concentric cross-sections each describable by its diameter. FIG. **4**B highlights the height HA**2** of the implant **100** in a deployed configuration measured between the inversions **122**, **124**, an outer diameter D**1** of the outer section **142**, which corresponds to the diameter of the open end **114**, and the outer diameter D**2** of the middle section **144**. It should be understood that the height HA**2** of example implant **100** can be varied to match the height HA**1** of spherical cavity **10** such that a proximal end of segment **110** is flush with a proximal end of the spherical cavity **10** in the deployed state. For example, the proximal end of segment **110** can be flush with the plane **18** when in the deployed configuration, as shown in FIG. **4**A. Although FIG. **4**B illustrates only one example deployed configuration, it should be understood that the height and diameter of example implants described herein **100** and portions thereof can be varied to fit spherical cavities **10** of varying dimensions.

FIG. **5** is a flowchart of a method **500** for constructing an endovascular treatment system. In block **502**, the method can include providing a segment **110** including an open end **114** and a proximal end **112**. The segment can include a braided tubular implant. The open end can be on a distal end of the segment.

In block **504**, the method can include shaping the segment to a predetermined shape to which the segment is capable of self-expanding. Shaping the segment to a predetermined shape can include inverting the segment to form a proximal inversion **122** folded towards the distal direction thereby defining an outermost section **142** of the segment. Shaping the segment to a predetermined shape can include inverting the segment to form a distal inversion **124** folded toward the proximal direction to define a middle section **144** between the proximal and distal inversion of the segment. The middle section of the segment can be at least partially surrounded by the outermost section and define an innermost section **146** between the distal inversion and the proximal end **112** that is at least partially surrounded by the middle section **144**.

In block **506**, the method can include providing a detachment feature **164** that is affixed to the proximal end of the segment. In some examples, the detachment feature **164** can include one or more balls **164**A that are configured to fit into one or more indentations **162**A formed within a distal end of push wire **160**. In some examples, the detachment feature **164** can include one or more wire loops **164**B formed from a proximal end of the segment **110** of the implant **100**, and the one or more wire loops **164**B are configured to fit into the one or more indentations **162**B formed within a distal end of push wire **160**.

In block **508**, the method can include providing a push wire **160** that includes one or more indentations **162** in a distal end of the push wire **160**.

In block **510**, the method can include recessing at least a portion of the detachment feature **164** into the one or more indentations on the push wire.

In block **512**, the method can include positioning a sleeve **170** at least partially surrounding the push wire **160** and the proximal end **112** of the segment **110**.

FIG. **6** is a flowchart of a method **601** for occluding a spherical cavity. In block **602**, the method can include providing an expandable segment **110** that includes an open end **114** and a proximal end **112** in a non-deployed configuration.

In block **604**, the method can include affixing the proximal end **112** to a push wire **160** via a detachment feature **164**.

In block **606**, the method can include positioning a sleeve over the push wire **160** and the detachment feature **164** to prevent detachment of the proximal end **112** from the distal end of the push wire **160**.

In block **609**, the method can include delivering the expandable segment **110**, push wire **160**, and sleeve **170** to a treatment site via a microcatheter **600**.

In block **610**, the sleeve **170** can be slid proximally, thereby releasing the proximal end **112** of the segment **110** from the distal end of the push wire **160**. In response, the segment **110** can expand to occlude the spherical cavity **10**.

The tubular braid **110** of the example implant **100** can include memory shape material that can be heat set to a predetermined shape, can be deformed for delivery through a catheter, and can self-expand to an implanted shape that is based on the predetermined shape and confined by the anatomy of the aneurysm in which it is implanted.

The example implant **100** described herein can rely on a radial outward force to anchor the implant within the sac of an aneurysm. To this end, the braid **110** can be shaped to a predetermined shape having a diameter that is greater than its height so that the braid is radially constricted when implanted in an aneurysm. The ratio of diameter to height of the braid **110** in a respective predetermined shape can be within the range of 2:1 to 1:3 to treat aneurysms of many known sizes and shapes.

The descriptions contained herein are examples of embodiments of the invention and are not intended in any way to limit the scope of the invention. As described herein, the invention contemplates many variations and modifications of the implant, including alternative materials, alternative geometries, alternative detachment features, alternative delivery systems, alternative means for forming a braid into a predetermined shape, alternative treatment methods, etc. These modifications would be apparent to those having ordinary skill in the art to which this invention relates and are intended to be within the scope of the claims which follow.

What is claimed is:

1. An endovascular implantation system, comprising:

an occlusive segment configured to occlude a spherical cavity and comprising an open end and a proximal end;

a push wire positioned proximal of the proximal end of the occlusive segment and comprising one or more indentations on an outer surface of the push wire approximate a distal end of the push wire;

a detachment feature affixed to the proximal end of the occlusive segment and shaped to at least partially recess into the one or more indentations of the push wire; and

a sleeve surrounding at least a portion of the push wire, wherein in a non-deployed configuration, the sleeve surrounds at least a portion of the detachment feature and thereby prevents release of the proximal end of the occlusive segment from the push wire, and

wherein in a deployed configuration, the sleeve is retracted proximally from the detachment feature to thereby detach the occlusive segment from the push wire.

2. The endovascular implantation system of claim **1**, wherein the detachment feature further comprises a plurality of wire loops extending from the proximal end of the occlusive segment that are at least partially recessed into the one or more indentations of the push wire in the non-deployed configuration.

3. The endovascular implantation system of claim **2**, wherein the one or more indentations comprise a plurality of laser-etched pockets, each of the laser-etched pockets shaped to receive a corresponding wire loop of the plurality of wire loops.

4. The endovascular implantation system of claim **1**, wherein the detachment feature further comprises a plurality of balls extending from the proximal end that are configured to fit into the one or more indentations of the push wire in the non-deployed configuration.

5. The endovascular implantation system of claim **1**, wherein in the deployed configuration, the occlusive segment expands to occlude the spherical cavity.

6. The endovascular implantation system of claim **1**, wherein in the deployed configuration the occlusive segment extends in a distal direction from the proximal end and comprises two inversions separating three sections which at least partially overlap each other such that the proximal end is affixed to an innermost section of the three sections, and a middle section of the three sections extends between the two inversions and is positioned within an outermost section and around the innermost section.

7. The endovascular implantation system of claim **1**, further comprising:

a microcatheter configured to deliver the occlusive segment, the push wire, and the sleeve to the spherical cavity in the non-deployed configuration, the microcatheter comprising a lumen,

wherein the segment comprises a diameter in the non-deployed configuration sized to fit within the lumen of the microcatheter.

8. The endovascular implantation system of claim **1**, further comprising a band approximate the proximal end of the occlusive segment, wherein the band comprises radiopaque material.

9. The endovascular implantation system of claim **8**, wherein, in the deployed configuration, the open end is positioned approximate a distal wall of the spherical cavity, and the band is suspended within the spherical cavity.

10. A method for constructing an endovascular implantation system, the method comprising:

providing an occlusive segment comprising an open end and a proximal end;

shaping, as follows, the occlusive segment to a predetermined shape to which the segment is capable of self-expanding:

inverting the segment to form a proximal inversion folded toward a distal direction thereby defining an outermost section of the segment, and

inverting the segment to form a distal inversion folded toward the proximal direction thereby defining a middle section between the proximal and distal inversion of the segment that is at least partially surrounded by the outermost section and defining an innermost section between the distal inversion and the proximal end that is at least partially surrounded by the middle section;

providing a detachment feature affixed to the proximal end of the segment;

providing a push wire comprising one or more indentations approximate a distal end of the push wire;

recessing at least a portion of the detachment feature into the one or more indentations of the push wire; and

positioning a sleeve to at least partially surround the distal end of the push wire and the detachment feature.

11. The method of claim **10**, wherein the sleeve is effective to retain the proximal end of the occlusive segment attached to the push wire.

12. The method of claim **10**, wherein the detachment feature further comprises a plurality of wire loops extending from the proximal end that are configured to fit into the one or more indentations positioned on the distal end of the push wire.

13. The method of claim **12**, wherein the one or more indentations comprise a plurality of laser-etched pockets, each of the laser-etched pockets shaped to fit a corresponding wire loop of the plurality of wire loops.

14. The method of claim **10**, wherein the detachment feature further comprises a plurality of balls extending from the proximal end that are configured to fit into the one or more indentations positioned on the distal end of the push wire.

15. A method of occluding a spherical cavity, the method comprising:

providing an expandable segment comprising an open end and a proximal end in a non-deployed configuration;

affixing the proximal end to a push wire via a detachment feature;

positioning a sleeve over the push wire and the detachment feature to prevent detachment of the proximal end from a distal end of the push wire;

delivering the expandable segment, push wire, and sleeve to a treatment site via a microcatheter; and

sliding the sleeve proximally, thereby releasing the proximal end from the distal end of the push wire and causing the expandable segment to expand to a deployed configuration to occlude the spherical cavity.

16. The method of claim **15**, wherein the detachment feature further comprises a plurality of wire loops extending from the proximal end that are configured to fit into one or more indentations positioned on a distal end of the push wire.

17. The method of claim **16**, wherein the one or more indentations comprise a plurality of laser-etched pockets, each of the laser-etched pockets shaped to fit a corresponding wire loop of the plurality of wire loops.

18. The method of claim **15**, wherein the detachment feature further comprises a plurality of balls extending from the proximal end that are configured to fit into the one or more indentations positioned on the distal end of the push wire.

19. The method of claim **15**, wherein in the deployed configuration, the proximal end of the expandable segment is flush with a proximal end of the spherical cavity.

20. The method of claim **19**, wherein in the deployed configuration the expandable segment comprises two inversions separating three sections which at least partially overlap each other such that the proximal end of the expandable segment is affixed to an innermost section of the three sections and a middle section of the three sections extends between the two inversions and is positioned within an outermost section and around the innermost section.

* * * * *